

OPO-Based Combinatorial Optimization

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- Combinatorial Optimization Problems

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- Coherent Ising Machines

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- Performance Challenges

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- Conclusion & Future Work

Combinatorial Problems

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- *Solutions* are *candidate solutions* that satisfy all given conditions.

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How hard is a given computational problem to solve?

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- *Time complexity of a problem Π* : Computation time required for solving a given instance π of Π using the most efficient algorithm for Π .
- *NP*: Class of problems that can be solved in polynomial time by a nondeterministic machine. *Note*: nondeterministic $\neq$ randomised; non-deterministic machines are idealised models of computation that have the ability to make perfect guesses.

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- *NP-complete*: Among the most difficult problems in NP; believed to have at least exponential time-complexity for any realistic machine or programming model.
- *NP-hard*: At least as difficult as the most difficult problems in NP, but possibly not in NP (i.e., may have even worse complexity than NP-complete problems).

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Recent work explores integrating analog physics-based hardware algorithms and hardware by mapping the cost function to the energy landscape of simple physical systems, including the classical **Ising model**.

Recall: The Ising Model

The Ising model is a *mathematical model* describing a lattice of spins $\vec{\sigma}$.

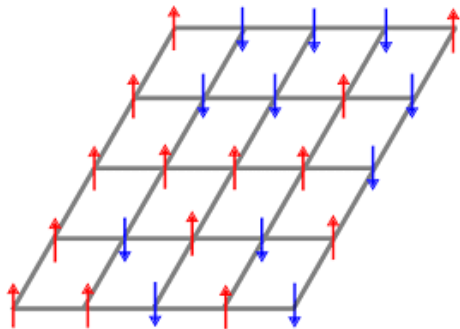


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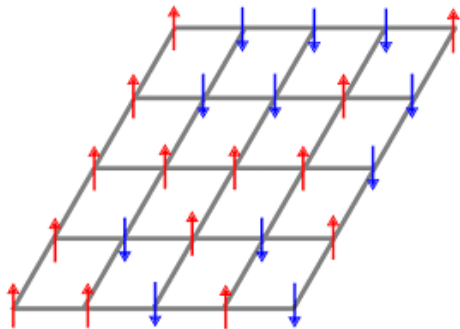


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- Each spin can be in one of two states: *spin-up* or *spin-down*, denoted by $\sigma_i = \pm 1$.
- The Hamiltonian, describing the energy of the system, is given by

$$H(\sigma) = - \sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j$$

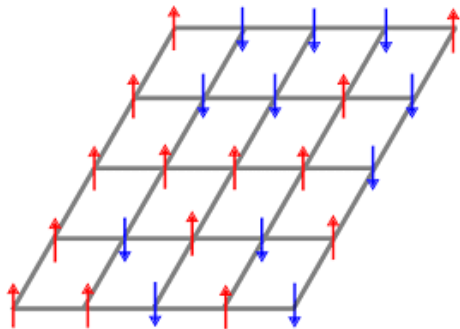


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Nonlinear Optics & Optical Parametric Oscillators

Linear Optics

- Light is deflected or delayed but its frequency is unchanged.

⇒ *Optics of weak light.*

Linear vs. Nonlinear Optics

Linear Optics

- Light is deflected or delayed but its frequency is unchanged.

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Nonlinear Optics

- We are concerned with the effects that light itself induces through the medium.

⇒ *Optics of intense light.*

Recall: Electric Dipole Moments

Electric dipole moments measure the system's *polarity*:

$$\mathbf{p} = q\mathbf{d}.$$

Here, the displacement vector $\mathbf{d}$ points from the negative charge magnitude $-q$ to the positive charge magnitude q .

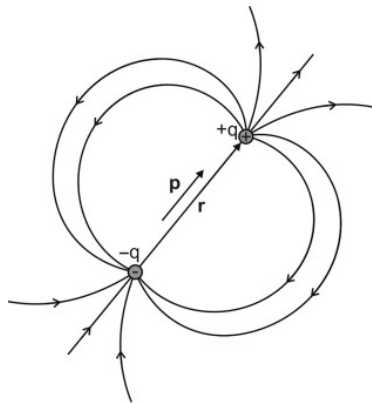


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While the molecules may remain neutral in charge, an **electric dipole moment** forms.

Polarization Density (Classical Electromagnetism)

The vector field expressing the volumetric density of permanent or induced electric dipole moments in a dielectric material.

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- (*scalar*) $\tilde{E}(t)$: The strength of an applied optical field.

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In contrast, **nonlinear optics** expresses the polarization $\tilde{P}(t)$ as a power series in the field strength $\tilde{E}(t)$:

$$\begin{aligned}\tilde{P}(t) &= \epsilon_0 [\chi^{(1)} \tilde{E}(t) + \chi^{(2)} \tilde{E}^2(t) + \chi^{(3)} \tilde{E}^3(t) + \dots] \\ &\equiv \tilde{P}^{(1)}(t) + \tilde{P}^{(2)}(t) + \tilde{P}^{(3)}(t) + \dots,\end{aligned}$$

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Note: We assume the polarization at time t depends only on the instantaneous value of the electric field.

We consider the optical field consisting of two distinct frequency components:

$$\tilde{E}(t) = E_1 e^{-i\omega_1 t} + E_2 e^{-i\omega_2 t} + \text{c.c.}$$

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Assuming $\tilde{P}^{(2)}(t) = \epsilon_0 \chi^{(2)} \tilde{E}(t)^2$, we find

$$\begin{aligned} \tilde{P}^{(2)}(t) &= \epsilon_0 \chi^{(2)} [E_1^2 e^{-2i\omega_1 t} + E_2^2 e^{-2i\omega_2 t} + 2E_1 E_2 e^{-i(\omega_1 + \omega_2)t} \\ &\quad + 2E_1 E_2^* e^{-i(\omega_1 - \omega_2)t} + \text{c.c.}] + 2\epsilon_0 \chi^{(2)} [E_1 E_1^* + E_2 E_2^*] \\ &= \sum_n P(\omega_n) e^{-i\omega_n t}. \end{aligned}$$

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Note: Summation extends over positive and negative frequencies ω_n .

The complex amplitudes of the frequency components of nonlinear polarization are:

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- Optical Rectification

$$P(0) = 2\epsilon_0 \chi^{(2)} [E_1 E_1^* + E_2 E_2^*]$$

Difference-Frequency Generation (DFG)

DFG is described by a nonlinear polarization of the form

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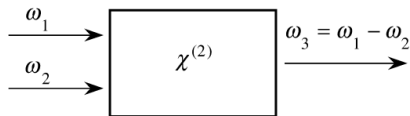


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The frequency of the generated wave is the difference of those of the applied fields!

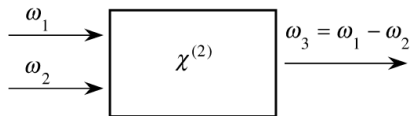


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Difference-Frequency Generation (DFG) - Continued

- 1 The atom absorbs a photon of frequency ω_1 and jumps to the highest virtual level.

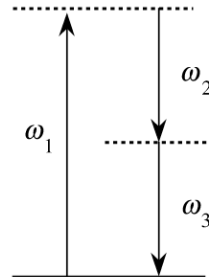


Figure: Energy-level description.

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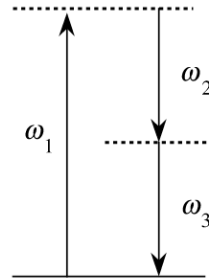


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Optical Parametric Amplifier (OPA)

- Conservation of energy requires that for every photon created at the difference frequency $\omega_3 = \omega_1 - \omega_2$, a photon at the higher input frequency ω_1 must be destroyed and a photon at the lower frequency ω_2 must be created.

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⇒ The lower-frequency input field is amplified by DFG! (*Optical Parametric Amplification (OPA)*)

Optical Parametric Oscillator (OPO)

Placing the nonlinear crystal inside an optical resonator amplifies the ω_2 and/or ω_3 fields

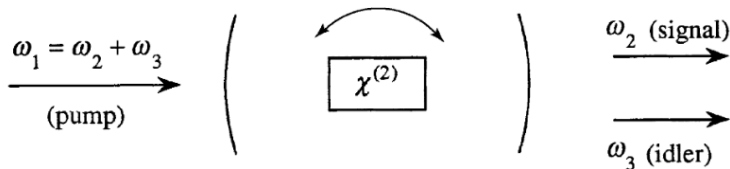


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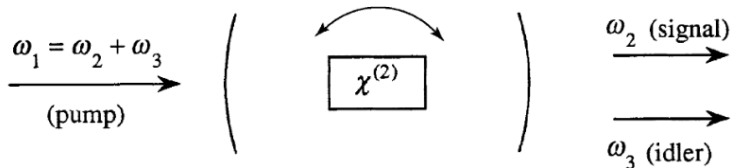


Figure: The optical parametric oscillator.

- The cavity end mirrors have high reflectivities at frequencies ω_2 and/or ω_3 .
- The output frequencies can be tuned by adjusting the orientation of the crystal.

Threshold

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- Occurs when the gain provided by the nonlinear interaction compensates for all the losses in the system.
- Ensures the parametric gain equals the round-trip losses within the optical resonator.

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- **Bistability:** When operating DOPOs above threshold, OPOs can oscillate in only one of two discrete phases: 0 or π relative to the pump laser.
- We can map Ising spins onto the bistability of their optical phase $\rightarrow$ solve combinatorial optimization problems!

Coherent Ising Machines

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It exploits the **bistability** of Degenerate Optical Parametric Oscillators: When operated above threshold, they oscillate in only one of two discrete phases: 0 or π relative to the pump laser.

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- z_i : coupling term.

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CIM - Mathematical Framework (Continued)

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- **J**: Ising coupling matrix ($J_{ii} = 0, J_{ij} = J_{ji}$);
- ξ : coupling strength acting as normalization constant for J_{ij} ($\xi > 0$).

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- **Freeze-out:** The domain evolution often freezes at a point t_f , beyond which the DOPO amplitudes remain unchanged.
- **Amplitude Heterogeneity:** DOPO amplitudes can deviate from their ideal values $\pm\sqrt{\alpha}$ (for target pump amplitude parameter α).

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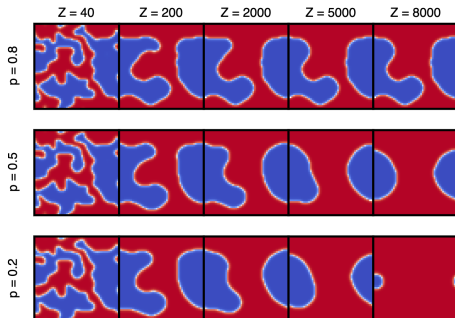
Note: We incorporate a checkerboard pattern into the DOPO amplitudes to visualize domain dynamics without influencing measurement outcomes.

Implications of Freeze-Out (Continued)

From a random initial spin state, we vary the normalized pump amplitude ($p = 0.2, 0.5, 0.8$), mimicking DOPOs below threshold.

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Two-Stage Process:

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- 2 **Saturation:** As pump energy decreases, DOPOs achieve maximum amplitude $\implies$ inward radial movement of domain walls (+ eventual disappearance).

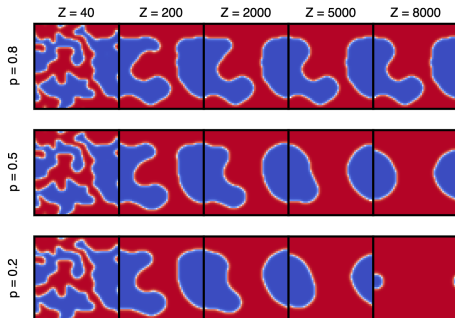


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- The speed of domain wall movement is influenced by *wall curvature* → more curved walls move faster.

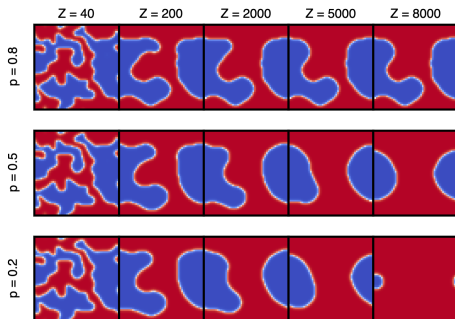


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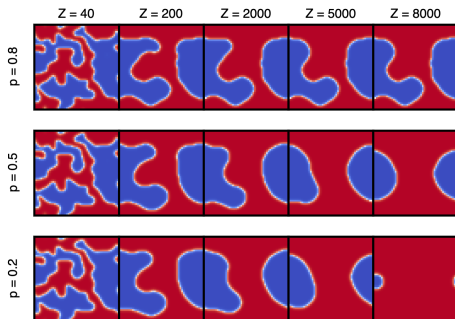


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- At low pump amplitude $p = 0.2$, the ground state is achieved.
- At high amplitudes, the system *freezes* (becomes trapped in a local minima).

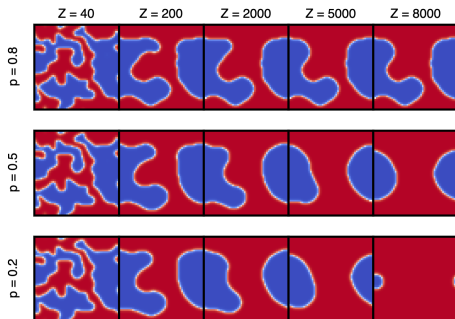


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Implications of Freeze-Out (Continued)

We analyze the time evolution of DOPO amplitudes for pump amplitudes $p = 0.2$ and $p = 0.8$.

- We clearly see the point where the DOPO amplitudes for $p = 0.8$ become trapped.

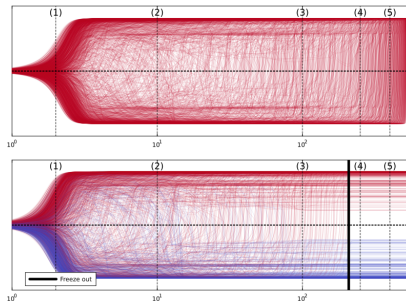


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⇒ The pump amplitude is critical in shaping the behavior of domains.

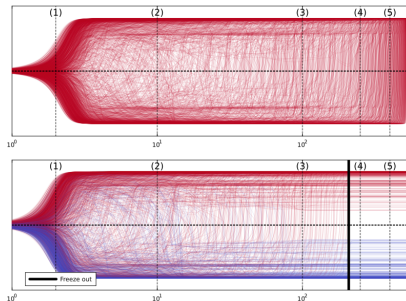


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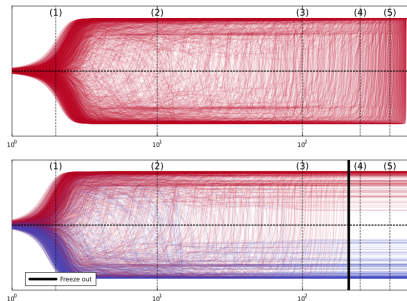


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- At low pump amplitude, energy decreases gradually until reaching the ground state.

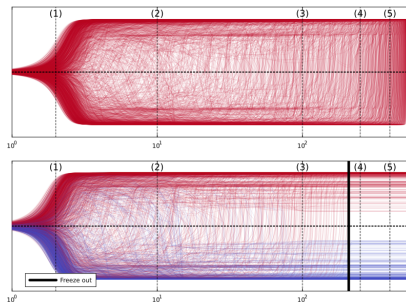


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- Trapping in local energy minima.
- High spin temperatures.

Trapping in Local Energy Mimima

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However, observations of freeze-out do not support such parallel domain wall formations → We can rule out local energy minima as a cause of freeze-out.

Trapping in Local Energy Mimima (Continued)

To understand the freeze-out process in CIMs further, we compare ground state searches to *Markov Chain Monte Carlo (MCMC)* simulations using the Metropolis-Hastings algorithm at varying temperatures.

Recall: Metropolis-Hastings Algorithm

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Note: Spins are updated sequentially, resembling injection pulse updates in the CIM.

Trapping in Local Energy Minima (Continued)

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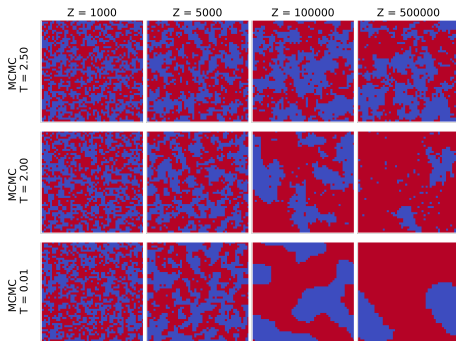


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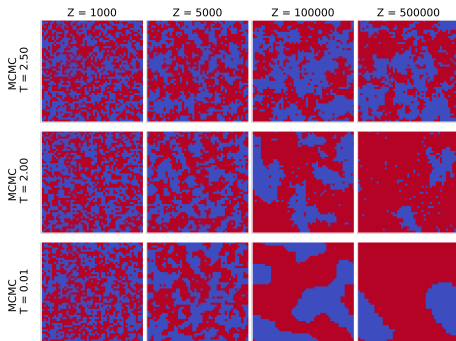


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⇒ CIMs mimic low-temperature MCMC behavior!

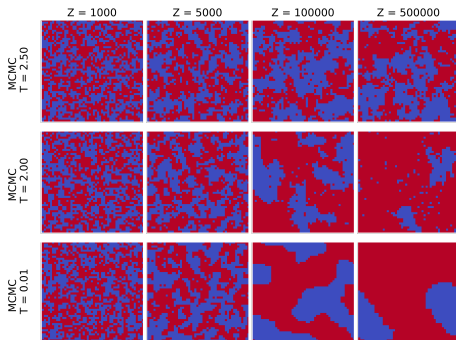


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Navigation icons: back, forward, search, etc.

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- Aims to achieve homogeneous OPO amplitudes and meet mapping conditions (above threshold).
- **But:** nonlinear filtering insufficient in preventing system from becoming trapped in local minima!
 - As problem size N increases, NP-hard Ising problems exhibit an exponentially increasing number of local minima.

Chaotic Amplitude Control (CIM - CAC)

We can destabilize local minima of the binary spin Hamiltonian by adding an auxiliary error pulse amplitude $e_i > 0$:

$$\begin{aligned}\frac{dx_i}{dt} &= -x_i^3 + (p-1)x_i + e_i z_i \\ \frac{de_i}{dt} &= -\beta(x_i^2 - \alpha)e_i.\end{aligned}$$

Here,

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Note: The system of equations exhibits inherent chaotic dynamics.

Alternatively, we consider CIM with chaotic feedback control:

$$\begin{aligned}\frac{dx_i}{dt} &= -x_i^3 + (p-1)x_i + e_i z_i \\ \frac{de_i}{dt} &= -\beta(z_i^2 - \alpha)e_i.\end{aligned}$$

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Note: We monitor the feedback signal z_i instead of internal amplitude x_i .

The final system we consider features distinct equations governing its dynamics:

$$\begin{aligned}\frac{dx_i}{dt} &= -x_i^3 + (p-1)x_i - f(cz_i) - k(z_i - e_i), \\ \frac{de_i}{dt} &= -\beta(e_i - z_i).\end{aligned}$$

Here:

- f : sigmoid-like, nonlinear function (e.g., $f(z) = \tanh(z)$);
- p, k, c, β : system parameters.

Separable Feedback Control (CIM - SFC) (Continued)

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'Separable' as it explicitly separates two essential aspects of CIM-CAC and CIM-CFC: amplitude heterogeneity and local minima destabilization!

- $f(cz_i)$ generates mutual coupling and addresses amplitude heterogeneity;
- $k(z_i - e_i)$ helps destabilize local minima and detects sharp changes in z_i .

Difference between CIM-CAC/CIM-CFC and CIM-SFC

CIM-CAC/CIM-CFC

- Error signal e_i is a strictly positive value that modulates the mutual coupling term exponentially.

CIM-SFC

- Error signal e_i stores sign information and spans the same range of values as the mutual coupling signal z_i .

To understand the similarities between CIM-CAC, CIM-CFC, and CIM-SFC, we analyze their fixed points by imposing:

$$\frac{dx_i}{dt} = 0, \quad \frac{de_i}{dt} = 0, \quad \forall i.$$

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since $\beta, e_i > 0$.

Fixed point analysis

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$$\frac{dx_i}{dt} \stackrel{!}{=} 0 = -x_i^3 + (p-1)x_i + e_i z_i \implies e_i = \frac{x_i^3 - (p-1)x_i}{\xi\{\mathbf{J}\vec{x}\}_i}.$$

Fixed point analysis (Continued)

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$$e_i = \frac{1 - \alpha - p}{\xi h_i \sigma_i} \implies |e_i| \propto \frac{1}{|h_i|},$$

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Note: The same relationship for CIM-CFC can be established.

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Note: Only valid if $f(cz_i)$ is odd function taking values of $+1$ for $cz \gg 1$ and -1 for $cz \ll 1$.

CIM-CAC/CIM-CFC

$$x_i \propto \sigma_i$$
$$|e_i| \propto \frac{1}{|h_i|}$$

CIM-SFC

$$x_i \propto \sigma_i$$
$$|e_i| \propto |h_i|$$

This difference makes CIM-SFC more robust to quantum noise from reservoirs and pump sources.

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- ① Similarities;
- ② Correlation properties;
- ③ Differences in trajectories.

Euler Approximation

For this, we approximate the time-evolution of the systems using a Euler approximation as follows:

$$x_i(t + \Delta t) = x_i + \Delta t \frac{dx_i}{dt},$$
$$e_i(t + \Delta t) = e_i + \Delta t \frac{de_i}{dt},$$

where Δt denotes the time step.

To understand the similarities between CIM-CAC/CIM-CFC and CIM-SFC, we will express CIM-CAC, CIM-CFC, and CIM-SFC as

$$\frac{dx_i}{dt} = -x_i^3 + (p-1)x_i - I_i(t).$$

Here:

- $M_i(t) = z_i(t) = \xi \sum_j J_{ij} x_j(t)$: Mutual-coupling signal.
- $I(t)$: Injection-feedback signal (depends on time evolution of $M_i(t)$).

Similarity Analysis (Continued)

CIM

$$I(t) = M(t)$$

CIM (Nonlinear Feedback)

$$I(t) = \tanh(M(t))$$

CIM (Chaotic Feedback)

$$I(t) = e(t)M(t), \text{ where}$$
$$\frac{de(t)}{dt} = -\beta e(t) (e(t)^2 M(t)^2 - \alpha)$$

CIM (Separable Feedback)

$$I(t) = \tanh(cM(t))$$
$$+ k (M(t) - e(t)), \text{ where}$$
$$\frac{de(t)}{dt} = -\beta (e(t) - M(t))$$



Similarity Analysis (Continued)

We examine how $I_i(t)$ varies w.r.t. $M_i(t)$:

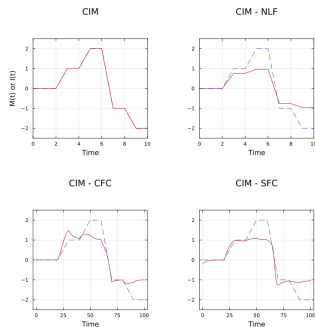


Figure: $M_i(t)$ (blue/dashed) and $I_i(t)$ (red/solid) in four different feedback systems.

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We examine how $I_i(t)$ varies w.r.t. $M_i(t)$:

- If $M_i(t)$ stays constant, $I_i(t)$ will converge to $\text{sign}(M_i(t))$.

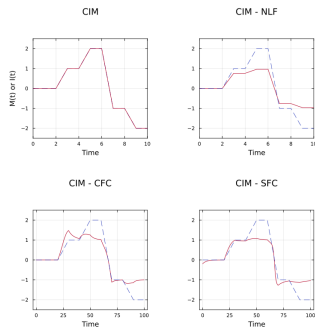


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- If $M_i(t)$ stays constant, $I_i(t)$ will converge to $\text{sign}(M_i(t))$.
- Sharp changes in $M_i(t)$ cause $I_i(t)$ to deviate from ± 1
 - Triggers destabilization of local minima and exploration of new spin configurations.

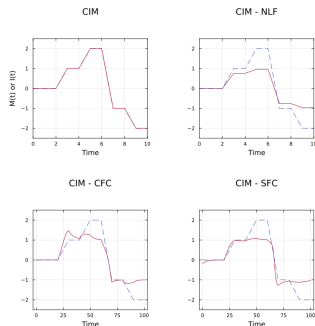


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Correlation Properties

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To demonstrate, consider correlation between $x_1 \sim \mathcal{N}(0, 0.25^2)$ and $x_2 = x_1 + 0.001Y$, where Y is standard Gaussian.

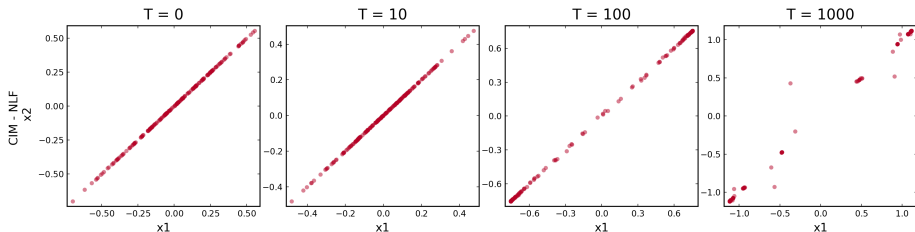


Figure: Signal pulse amplitude correlations at different evolution time in CIM-NLF

Correlation Properties (Continued)

We analyze the presence of this property across CIM-CAC, CIM-CFC, and CIM-SFC

- CIM-CAC and CIM-CFC show chaotic behavior, with trajectories diverging after approx. 100 steps
 - Sensitive to initial conditions.
- CIM-SFC shows nearly identical trajectories after 1000 iterations.

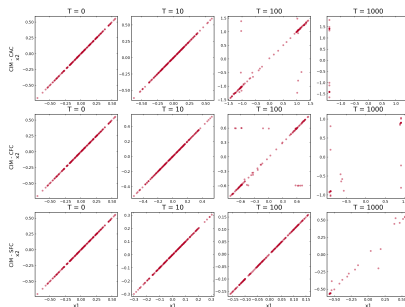


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Differences in Trajectories

We examine the trajectories of CIM-CAC, CIM-CFC, and CIM-SFC when finding the ground state of a 200×200 Ising lattice with perfect antiferromagnetic coupling between nearest neighbours.

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We consider both *fixed* hyperparameters, and *slowly modulated* hyperparameters.

Differences in Trajectories (Continued)

Fixed Parameters:

- CIM-SFC and CIM-CFC become trapped in stable periodic attractors.
- CIM-CAC finds a local minimum.

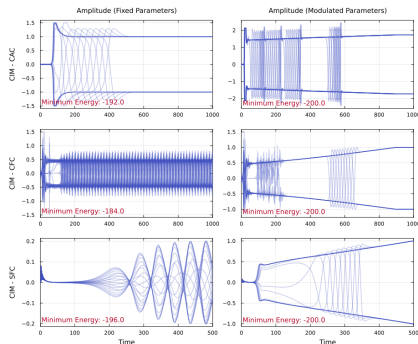


Figure: Trajectories of CIM-CAC, CIM-CFC, and CIM-SFC with fixed and modulated parameters.

Differences in Trajectories (Continued)

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Modulated Parameters:

- Slow modulation yields better results on average!
 - Reduces likelihood of systems becoming trapped in local minima and exhibiting oscillatory behavior.

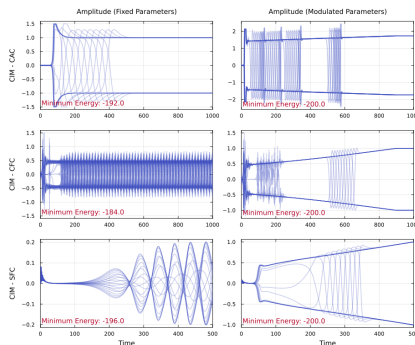


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Why Does Modulation Yield Better Results?

The average residual energy visited by CIM-CAC/CIM-CFC can be approximated as

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where constant K depends on problem type and size.

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where constant K depends on problem type and size.

We gradually reduce the system temperature by increasing p and α to produce an **annealing effect**.

Experiments

The MAX-CUT Problem

To compare CIM-CAC, CIM-CFC, and CIM-SFC, we consider the NP-hard MAX-CUT problem:

The MAX-CUT Problem

Consider an undirected graph $G = (V, E)$ with edge weight function $w : E \rightarrow \mathbb{R}$, given edges E and vertices V .

We aim to find a cut $(S, V \setminus S)$ that maximizes the sum of the weights of the edges crossing the cut over all possible cuts.

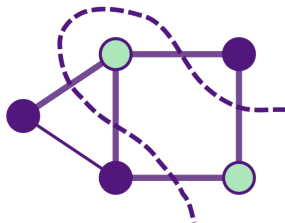


Figure: Illustration of the MAX-CUT Problem.

The MAX-CUT Problem (Continued)

Concretely, we wish to solve:

$$\begin{aligned}\max(w(S)) &= \max \left(\sum_{j \in S, l \in V \setminus S} w_{jl} \right) \\ &= \max \left(\frac{1}{4} \sum_{j, l \in V} w_{jl} - \frac{1}{4} \sum_{j, l \in V} w_{jl} \sigma_j \sigma_l \right).\end{aligned}$$

Here:

- $w_{jl} = w_{lj}$ if $(j, l) \in E$ and $w_{jl} = 0$ otherwise.
- $\sigma_j = +1$ if j -th vertex in S and $\sigma_j = -1$ otherwise.

The MAX-CUT Problem (Continued)

By setting the coupling coefficient J_{ij} in the Ising model to $-w_{jl}$ we translate the MAX-CUT problem to a ground state search problem of the Ising Hamiltonian!

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We will benchmark MAX-CUT performance using 6 G-Set instances¹ of size 800, covering random, toroidal, and planar graphs with varying edge densities and edge weights of +1, 0, and -1.

¹The G-Set instances are available at <https://web.stanford.edu/~yyye/yyye/Gset/>.



Parameter Choice - Coupling Strength ξ

We choose the same ξ constant for CIM-CAC, CIM-CFC, and CIM-SFC:

$$\xi = \sqrt{\frac{2N}{\sum J_{ij}^2}}, \quad (0.1)$$

This is because CIM-SFC needs $\tanh(cz_i)$ to shift from 'soft-spin' ($cz_i \approx 0$, $\tanh(cz_i) \approx cz_i$) to 'discrete-spin' ($|cz_i| \gg 0$ and $\tanh(cz_i) \approx \sin(cz_i)$).

- Ensures that z_i will be approx. $\sqrt{2}$ for a randomly chosen spin configuration.

Parameter Choice (CIM-CAC)

- We constrained the x_i variables to the range $[-1.5\sqrt{\alpha}, 1.5\sqrt{\alpha}]$ at each step.
- Algorithm parameters are linearly adjusted over T_r steps and then held constant for T_p steps.
- $x_i(0) \sim \mathcal{N}(0, 10^{-4})$ and $e_i(0) = 1 \forall i$.
- $\alpha : 1.0 \rightarrow 3.0$; $\beta : 0.3$ fixed.

Graph Type	N	Edge Weight	G-Set Instance	p	ΔT	T_r	T_p
Random	800	$\{+1\}$	1, 2	$-0.5 \rightarrow 1.0$	0.075	6000	666
Toroidal	800	$\{+1, -1\}$	11, 12	-4.0	0.1	4500	500
Planar	800	$\{+1\}$	14, 15	-1.0	0.05	18000	2000

Figure: Parameters for solving MAX-CUT instances on the G-Sets using CIM-CAC.

Parameter Choice (CIM-CFC)

- We constrained the x_i variables to the range $[-1.5, 1.5]$ at each step.
- Algorithm parameters are linearly adjusted over T_r steps and then held constant for T_p steps.
- $x_i(0) \sim \mathcal{N}(0, 0.1)$ and $e_i(0) = 1 \forall i$.
- $\alpha : 1.0$ fixed; $\beta : 0.15$ fixed.

Graph Type	N	Edge Weight	G-Set Instance	p	ΔT	T_r	T_p
Random	800	$\{+1\}$	1, 2	$-1.0 \rightarrow 1.0$	0.125	3600	400
Toroidal	800	$\{+1, -1\}$	11, 12	$-3.0 \rightarrow -1.0$	0.25	1800	200
Planar	800	$\{+1\}$	14, 15	$-2.0 \rightarrow 0.0$	0.125	7200	800

Figure: Parameters for solving MAX-CUT instances on the G-Sets using CIM-CFC.

Parameter Choice (CIM-SFC)

- No restriction of the x_i variables needed.
- Algorithm parameters are linearly adjusted over N steps.
- $x_i(0) \sim \mathcal{N}(0, 0.1)$ and $e_i(0) = 0 \forall i$.
- $\alpha : -1.0 \rightarrow 1.0$.

Graph Type	N	Edge Weight	G-Set Instance	c	β	k	ΔT	N Steps
Random	800	$\{+1\}$	1, 2	$1.0 \rightarrow 3.0$	$0.3 \rightarrow 0.0$	0.2	0.15	2666
Toroidal	800	$\{+1, -1\}$	11, 12	1.4	$0.05 \rightarrow 0.0$	0.32	0.4	2500
Planar	800	$\{+1\}$	14, 15	$-2.0 \rightarrow 0.0$	$0.05 \rightarrow 1.0$	0.2	0.125	5000

Figure: Parameters for solving MAX-CUT instances on the G-Sets using CIM-SFC.

Results

We generate 200 initial spin configurations and solve using the algorithms:

ID	N	Graph Type	Edge Weight	C_{OPT}^*	C_{CAC}^*	C_{CFC}^*	C_{SFC}^*
1	800	Random	$\{+1\}$	11624	11114	10926	10730
2	800	Random	$\{+1\}$	11620	11042	11014	10732
11	800	Toroidal	$\{+1, -1\}$	564	440	440	432
12	800	Toroidal	$\{+1, -1\}$	556	438	440	424
14	800	Planar	$\{+1\}$	3064	2713	2704	2235
15	800	Planar	$\{+1\}$	3050	2657	2695	2330

Figure: Best MAX-CUT value obtained using CIM-CAC, CIM-CFC, and CIM-SFC.

Observations

- No algorithm finds the optimal solution.
- CIM-CAC usually performs best.
- CIM-SFC usually performs weakest.

Results (Continued)

We generate 200 initial spin configurations and solve using the algorithms:

ID	N	Graph Type	Edge Weight	C_{OPT}^*	C_{CAC}^*	C_{CFC}^*	C_{SFC}^*
1	800	Random	$\{+1\}$	11624	11114	10926	10730
2	800	Random	$\{+1\}$	11620	11042	11014	10732
11	800	Toroidal	$\{+1, -1\}$	564	440	440	432
12	800	Toroidal	$\{+1, -1\}$	556	438	440	424
14	800	Planar	$\{+1\}$	3064	2713	2704	2235
15	800	Planar	$\{+1\}$	3050	2657	2695	2330

Figure: Best MAX-CUT value obtained using CIM-CAC, CIM-CFC, and CIM-SFC.

Potential Issues:

- Parameter Modulation not well tuned.
- Insufficient number of initial spin configurations generated.
- Not enough iterations.

Conclusions & Future Work

- The mechanisms have potential to tackle NP-hard optimization problems.
- CIM-CAC and CIM-CFC demonstrate chaotic behaviour and high sensitivity to initial conditions, potentially allowing for a more thorough exploration of the solution space.
- CIM-SFC typically converges to stable periodic orbits, maintaining deterministic behavior under most conditions.

- Further research needed to understand the parameter dependencies of these algorithms and optimize their performance across a broader range of problem types and sizes.
 - Novel work uses state-of-the-art reinforcement learning techniques to train agents that can determine optimal parameters based on the specifics of the problem instance.
- Valuable to explore whether incorporating prior information into the initial spin configuration could improve performance.

Thank You For Listening!

